# LELME2150 Thermal Cycles - Homework 2

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## 1 Model and main results

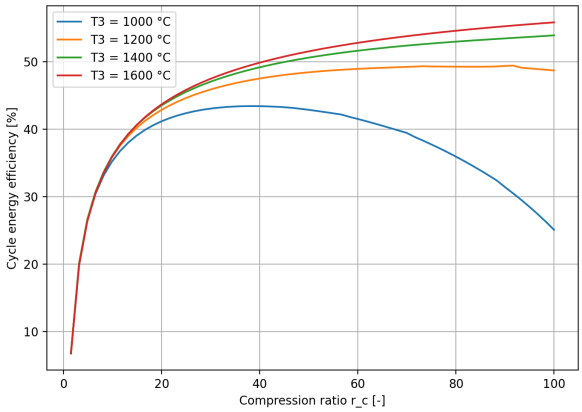
We modelled a gas turbine (compressor, combustion chamber, turbine) with energy and exergy balances, fuelled by methane. The model uses polytropic efficiencies, a pressure loss in the combustion chamber, mechanical losses, and temperature-dependent $c_p$ and $\gamma$ computed from the mixture composition (iterative procedure, see Appendix A). Table 1 gives the states for the reference case ($r_C = 18$, $T_3 = 1400$ °C).

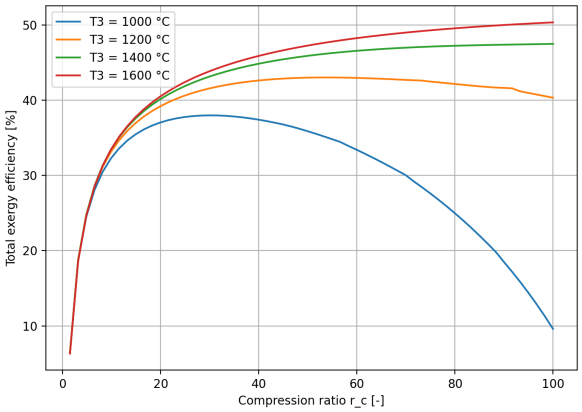
Table 1: State properties for the reference case.

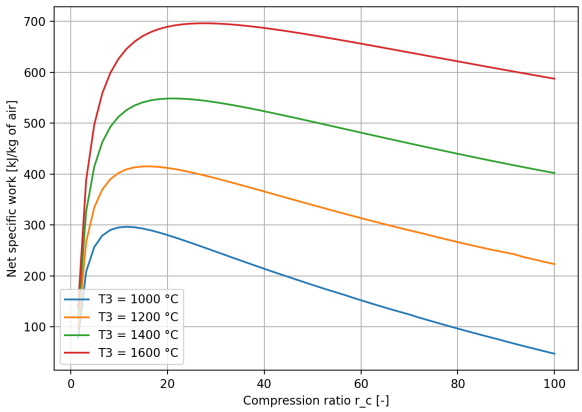
| **State** | $p$ [bar] | $T$ [K] | $h$ [kJ/kg] | $s$ [kJ/kgK] | $e$ [kJ/kg] |
|---|---|---|---|---|---|
| 1 | 1 | 288.15 | $\approx 0$ | $\approx 0$ | $\approx 0$ |
| 2 | 18 | 703.15 | 431.93 | 0.09 | 406.35 |
| 3 | 17.1 | 1673.15 | 1674.83 | 1.11 | 1324.84 |
| 4 | 1 | 928.51 | 720.42 | 1.19 | 346.05 |

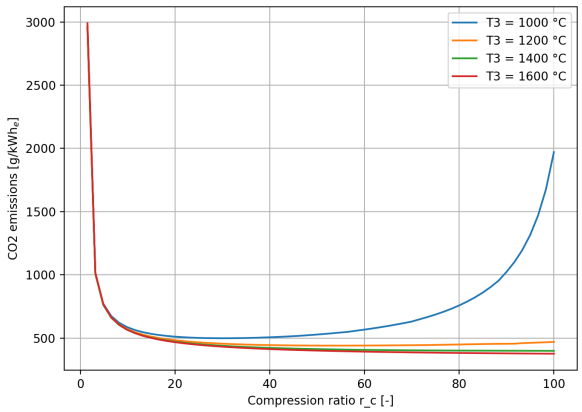
You can also see the Appendix B if you want to know how we've used the AI for this homework.

## 2 Parametric analysis 1 : compression ratio and turbine inlet temperature



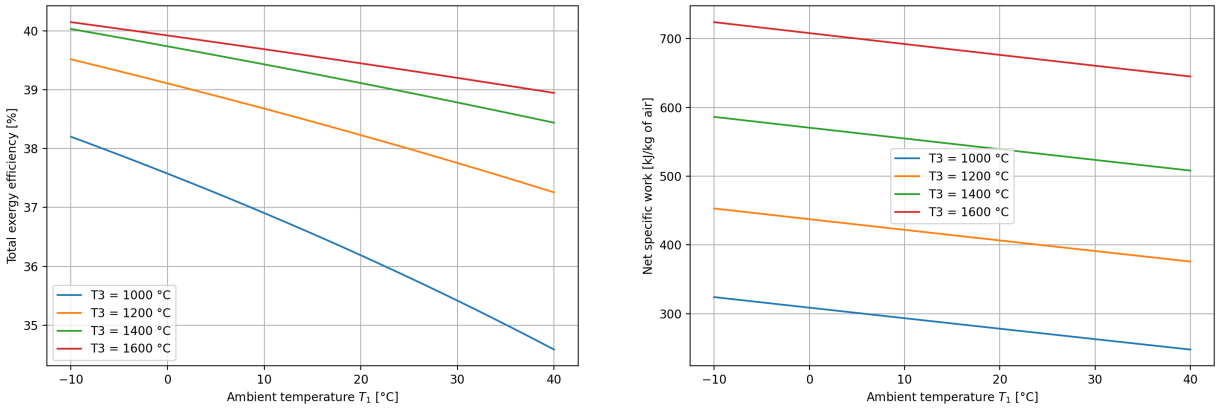




We observe that both efficiencies[1] and the net specific work go through a maximum whose value increases with $T_3$. This behaviour is explained by the trade-off between the thermodynamic efficiency gain with $r_c$ and the increase in compression work, which reduces the available net work. Since the net specific work peaks at a lower $r_c$ than the efficiency, the choice of the compression ratio results from a compromise between machine compactness and fuel consumption. The specific $CO_2$ emissions logically follow the inverse trend of the efficiency[2].

[1] In the following analysis, we only present one of the two efficiencies, since their behaviours are very similar.

[2] This behaviour holds for all the following analyses exepted when the fuel is changed, so it will not be mentioned again.

# 3 Parametric analysis 2 : Inlet air temperature difference




Both the total exergy efficiency and the net specific work decrease almost linearly as the ambient temperature $T_1$ increases. Since the turbine work is mainly set by $T_3$, which is fixed, while the compressor work is proportional to $T_1$, a warmer inlet air increases the work consumed by the compressor and reduces the net work available. The efficiency decreases accordingly. This explains why gas turbines lose performance in hot weather.

# 4 Parametric analysis 3 : fuel composition

Table 2: Performance for different fuels ($r_C = 18$, $T_3 = 1400$ °C, $P_e = 230$ MW).

| **Fuel** | $\lambda$ [-] | $\eta_{cyc}$ [%] | $\eta_{ex}$ [%] | $\dot{m}_f$ [kg/s] | $w_{net}$ [kJ/kg] | $CO_2$ [g/kWh] |
|---|---|---|---|---|---|---|
| Methane | 2.28 | 42.5 | 39.3 | 11.22 | 547 | 482 |
| Propane | 2.35 | 42.3 | 38.5 | 12.17 | 537 | 570 |
| Hydrogen | 2.69 | 43.5 | 42.6 | 4.54 | 574 | 0 |

The three fuels give similar energy efficiencies (42–44%), since $T_3$ and $r_C$ are the same. Hydrogen is slightly better, as its combustion products are rich in water vapour, which has a higher $c_p$ and increases the turbine work (574 against 547 kJ/kg of air for methane). The difference is larger for the exergy efficiency (42.6% against 39.3%), because the chemical exergy of hydrogen is close to its heating value, whereas it is higher than the heating value for hydrocarbons. Hydrogen also requires about 2.5 times less fuel mass flow, thanks to its much higher heating value per kg. Propane is the worst option, with 18% more $CO_2$ than methane, because of its higher carbon content per unit of energy. Hydrogen emits no $CO_2$ at the stack, but this does not account for its production.

# 5 Conclusion and suggested improvements

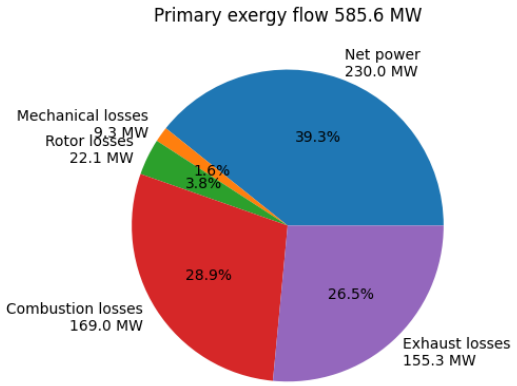



Figure 3: Exergy losses of the reference case.

The analysis shows that the compression ratio should be chosen as a compromise between compactness (maximum net work) and fuel consumption (maximum efficiency), and that performance is strongly affected by the ambient temperature. Several improvements can be considered:

- **Heat recovery** from the exhaust gases (regenerator or combined cycle), since the exhaust still carries a large amount of exergy, as seen in Figure 3.
- **Inlet air cooling** (e.g. evaporative cooling) to reduce the compressor work in hot weather, which is especially interesting for moderate $T_3$.
- **Increasing** $T_3$, limited by materials and blade cooling, to raise both efficiency and net work.
- **Intercooling** between compressor stages to reduce the compression work at high $r_c$.

The choice of the fuel also directly affects $CO_2$ emissions, fuels with higher carbon content produce more $CO_2$ per unit of energy released, while hydrogen combustion produces no direct $CO_2$ emissions.

# A Model details

Specific heats $c_p$ and $c_v$ are not constant: they are computed from the molar composition and molar masses of the air and combustion products. The ratio $\gamma(T)$ is therefore temperature dependent. During compression and expansion, $\gamma$ is updated iteratively (with a maximum number of iterations) to obtain an outlet temperature consistent with the prescribed polytropic efficiency.

# B Use of AI

In this report, we used a little bit of AI, for :

- Improving the presentation of certain sections
- Conducting supplementary research of a few small topics

We used them as a support. We remain responsible for the entire work (content, code, calculations, interpretations, and conclusions that are presented here).